



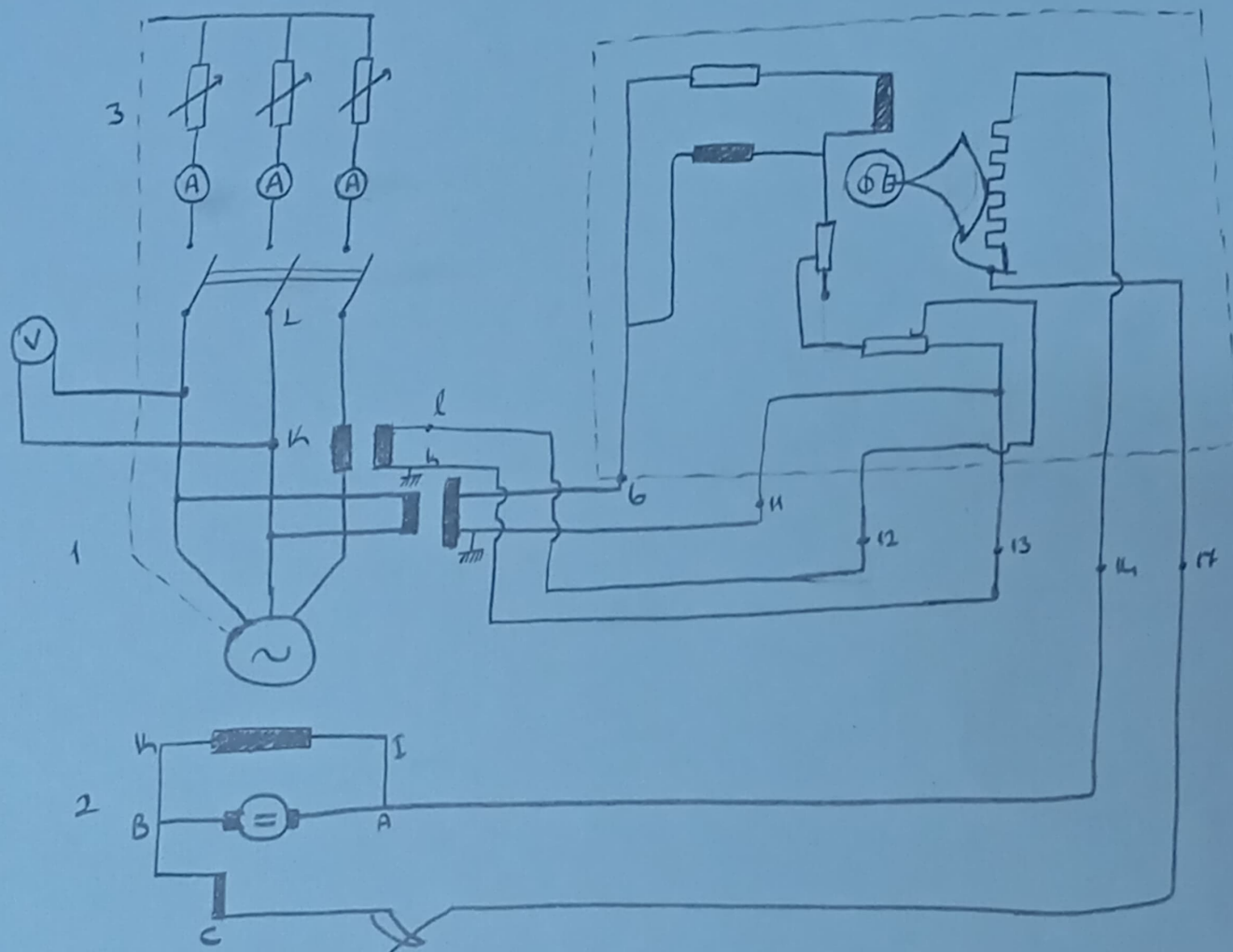
ITU Electrical-Electronics Faculty – Electrical Engineering Department

ELECTRICAL DISTRIBUTION AUTOMATION LABORATORY

LAB REPORT

Experiment No and Title	7: Voltage Regulation in Three-Phase Generators and Regulators
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TA responsible for the experiment	
Experiment date	12.03.2025
Submission date of the lab report	19.03.2025
Grade	

Connection Scheme



The objective of this experiment is to compare the variation of generator terminal voltage with and without the Brown-Boveri Regulator.

The regulator is designed to control the excitation of a synchronous generator and maintain a stable voltage despite load variations. Generator voltage depends on the strength of the rotating magnetic field, which is controlled by DC excitation current which comes from the exciter dynamo. Its output is controlled by the Brown-Boveri Regulator to regulate the main generator's voltage. Stronger field current means higher generator output voltage. Regulator adjusts the resistance in the exciter circuit to control the excitation current. This keeps the generator voltage stable even as loads change.

1) Why do we need voltage regulation?

Without proper voltage regulation, fluctuations in voltage can cause significant problems such as malfunctioning of electrical devices, overheating, inefficiency in power transmission, synchronization problems in power grids with multiple generators. For instance, renewable energy sources have fluctuating outputs due to environmental conditions. Voltage regulations ensure that these sources integrate smoothly with the main grid.

2) Explain the effect of armature reaction on generator voltage.

Armature reaction is the impact of the armature (stator) current on the magnetic field of the generator. In a synchronous generator, voltage is induced by the rotating magnetic field of the rotor. However, when a load is connected, the armature winding carries current, which produces its own magnetic field. This field interacts with the main rotor field, altering the net magnetic field inside the generator. This interaction leads to voltage drop or distortion in the generated voltage.

In an inductive load, the armature current lags behind the generator voltage. This produces a demagnetizing effect, meaning the armature reaction opposes the main field and reduces its strength.

In a capacitive load, the armature current leads the generator voltage. The armature reaction now has a magnetizing effect, meaning it strengthens the main field. But the generator voltage increases, possibly beyond safe limits.

We need to increase excitation current to compensate for the weakened field and reduce the excitation current to avoid excessive voltage rise.

3) How does a voltage regulator regulate voltage output of a generator? Explain the importance of regulators.

By controlling the excitation current. When load increases more current is drawn from the generator and voltage drops due to increased armature reaction and impedance. Regulator detects drop and increases excitation current. Stronger magnetic field boosts the generator voltage back to normal. When load decreases less current is drawn, voltage rises and regulator reduces excitation current.

Also, voltage regulators ensure balanced power sharing when multiple generators run in parallel. They also prevent dangerous voltage fluctuations.

4) Explain the regulator characteristics.

Proportional regulation, adjusts excitation gradually based on voltage deviation, small steady-state error remains.

Integral regulation adjusts excitation continuously to maintain exact voltage and there is no steady-state error.

excitation current

$$I_f = \frac{V_f \rightarrow \text{exciter voltage}}{R_f \rightarrow \text{exciter field resistance}}$$
$$E = k \cdot N \cdot \Phi$$

flux (is proportional to I_f)

$$\Phi \propto I_f$$

$$E \propto I_f \propto \frac{1}{R_f}$$

Higher $R_f \rightarrow$ Lower $I_f \rightarrow$ Lower E
Lower $R_f \rightarrow$ Higher $I_f \rightarrow$ Higher E

5) What should be the regulator characteristics of the parallel connected generators, so that reactive power is properly distributed among the generators?

Reactive power flows based on voltage differences between generators. Proportional regulators adjust voltage gradually as load changes. They also allow dynamic sharing of reactive power without large circulating currents.

Integral regulators try to keep voltage exactly constant, regardless of the load.

This causes one generator to take full voltage control, leading to excessive reactive current flow between generators. Therefore proportional regulator would be the ideal setup.

Since the BBR used in this experiment has proportional characteristics it can be used for parallel-connected generators.

6) How the load is distributed among two generators when one is equipped with integral characteristic regulator and the other is connected with proportional characteristic regulator?

Let's assume we have generator A with integral regulator and generator B with proportional regulator. As the system load increases B reduces its terminal voltage slightly. This causes A to take on most of the reactive power load. A is now overloaded with reactive power while B takes very little.

As the load decreases B's voltage increases slightly. A still maintains a constant voltage. A dominates the reactive power supply while B does very little.

References

1. Elektrik Testleri Laboratuvar Dersleri, Prof. Dr. Mustafa Beyram, 1999.
2. Power System Operation and Control, S. Sivaraman, G. Srinivasan, Pearson.
3. Power System Analysis, Stevenson W.D., 1994
4. Electric Machinery Fundamentals, Chapman S.J.

Regulators

Protocol Paper

This protocol paper will be filled in completely and delivered to the research assistant at the end of the experiment.

Date Of The Experiment	Lab. Session Day And Hour	Exp. Group
12.03.2025	Wednesday 12.30	6

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Yüksüz Durumda $f = 50 \text{ Hz}$
regülatör = 1.5

Load (A)	V_g	Reg konumu	Frekans [Hz]
1A	172V	1.4	50 Hz
1.5	171V	1.35	49.8 Hz
2	170V	1.3	49.6 Hz
2.5	168V	1.25	49.3
3	167V	1.15	49.0
3.5	166V	1.1	48.6
4A	164V	1.05	48.2

$$V = 175V$$

Yüksüz durumda $185V$
 $f = 50.1$

Load (A)	V_g	Frekans
1A	176	50.1
1.5A	162	50.1
2	168	50.2
2.5	132	50.1
3	109	50.2
3.5	91	50.2
4	55	

Generator Output Voltage vs Load Current

